# USER INTERFACE DESIGN

## EXPERIMENT 3

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**Difference between CLI(Command line interface) and GUI(Graphical user interface) and VUI(voice user interface)**

### 1.Command Line Interface

A Command Line Interface (CLI) allows the user to interact with the system by typing commands and inputs through the keyboard.

It is simple and fast because the user directly enters data without using buttons or graphics.

CLI programs require less memory and system resources compared to graphical applications.

In this experiment it creates the file and open the file. If the file is created then shows this file already exist.

### IMPLEMENTATION

```
import sys
import os

def create_file(filename):
    try:
```

```
        if os.path.exists(filename):
            print(f"File '{filename}' already exists.")
        else:
            with open(filename, 'w') as file:
                file.write("File created successfully!\n")
            print(f"File '{filename}' created successfully.")
    except Exception as e:
        print(f"An error occurred: {e}")

if __name__ == "__main__":
    if len(sys.argv) != 2:
        print("Usage: python create_file_cli.py <filename>")
    else:
        create_file(sys.argv[1])
```

**OUTPUT:**

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\ABISHAA\Downloads\UID> python cli.py abishaa.txt
File 'abishaa.txt' created successfully.
PS C:\Users\ABISHAA\Downloads\UID> python cli.py abishaa.txt
File 'abishaa.txt' already exists.
PS C:\Users\ABISHAA\Downloads\UID>
```

## 2.Graphical User Interface

A Graphical User Interface (GUI) allows users to interact with the system using visual elements like windows, buttons, and text boxes instead of typing commands.

It is user-friendly and easy to understand because users can click buttons and enter data in input fields.

GUI provides a better visual experience and is suitable for beginners who may not know command-line instructions.

In this experiment it creates the file and open the file. If the file is created then shows this file already exist.

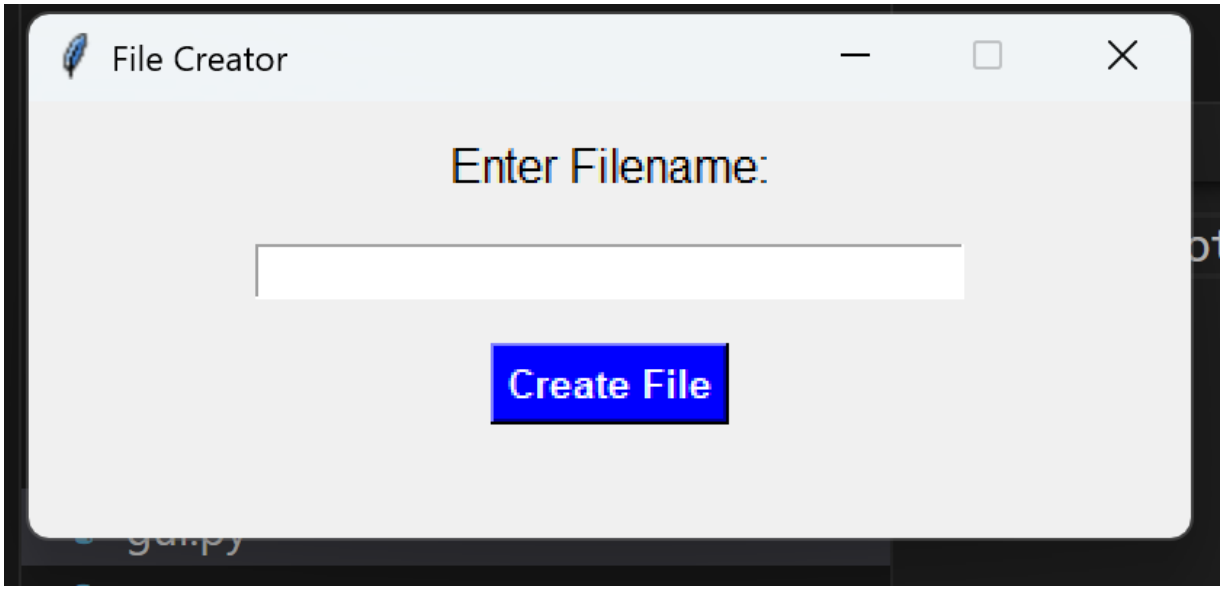
**IMPLENTATION:**

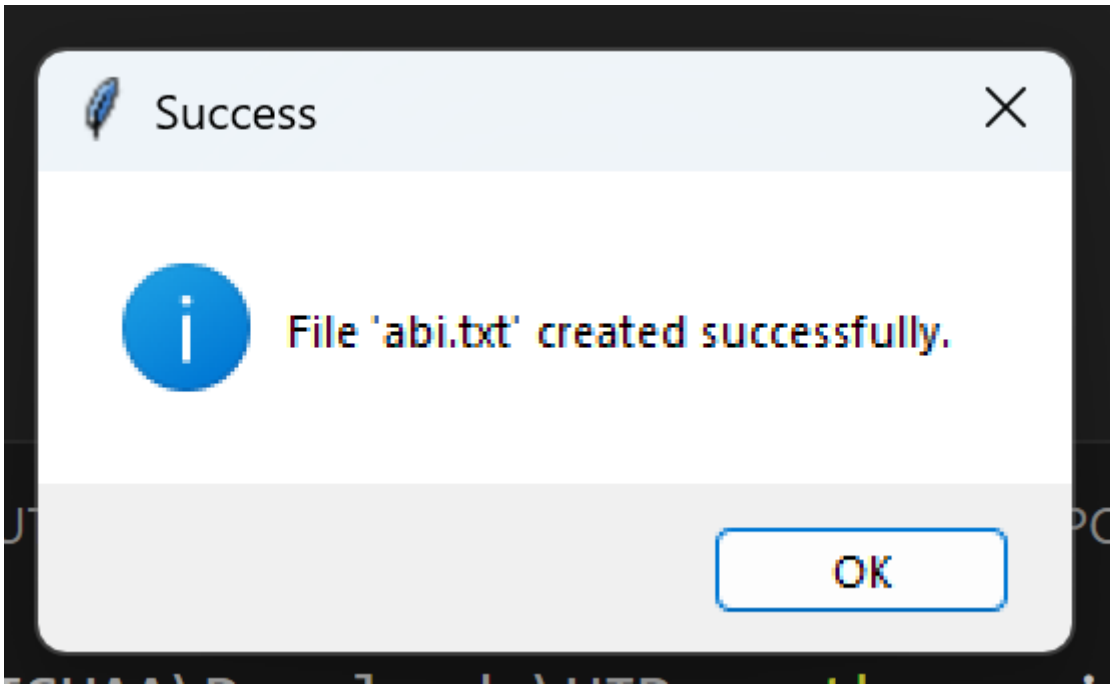
```
import tkinter as tk
from tkinter import messagebox
import os


def create_file():
    filename = filename_entry.get().strip()


    if not filename:
        messagebox.showwarning("Warning", "Please enter a filename!")
        return
```

```
    if os.path.exists(filename):
        messagebox.showinfo("Info", f"File '{filename}' already
exists.")
        return

    try:
        with open(filename, 'w') as file:
            file.write("File created successfully!\n")

        messagebox.showinfo("Success", f"File '{filename}' created
successfully.")
    except Exception as e:
        messagebox.showerror("Error", f"An error occurred: {e}")

root = tk.Tk()
root.title("File Creator")
root.geometry("400x150")
root.resizable(False, False)

tk.Label(root, text="Enter Filename:", font=("Arial",
12)).pack(pady=10)

filename_entry = tk.Entry(root, width=40)
filename_entry.pack(pady=5)
```

```
tk.Button(root, text="Create File",
        bg="blue",
        fg="white",
        font=("Arial", 10, "bold"),
        command=create_file).pack(pady=10)

root.mainloop()
```

**OUTPUT:**






**3.Voice User Interface**

**IMPLEMENTATION**

```
import speech_recognition as sr
import os

def create_file_from_voice(command):
    words = command.lower().split()

    if "create" in words and "file" in words:
```

```
        try:
            filename = words[words.index("file") + 1]

            if os.path.exists(filename):
                print(f"File '{filename}' already exists.")
            else:
                with open(filename, "w") as f:
                    f.write("File created using voice command.\n")

                print(f" File '{filename}' created successfully.")

        except IndexError:
            print("Please say: Create file filename.txt")

    else:
        print("Invalid command. Say: Create file filename.txt")

def listen():
    recognizer = sr.Recognizer()
    mic = sr.Microphone()

    print(" Listening... Say: Create file filename.txt")

    with mic as source:
        recognizer.adjust_for_ambient_noise(source)
```

```
        audio = recognizer.listen(source)

    try:
        command = recognizer.recognize_google(audio)
        print("You said:", command)
        create_file_from_voice(command)

    except sr.UnknownValueError:
        print("Sorry, could not understand audio.")
    except sr.RequestError:
        print("Speech Recognition service error.")

if __name__ == "__main__":
    listen()
```

**OUTPUT:**

```
PS C:\Users\ABISHAA\Downloads\UID> python vui.py
 Listening... Say: Create file filename.txt
Sorry, could not understand audio.
PS C:\Users\ABISHAA\Downloads\UID> python vui.py
 Listening... Say: Create file filename.txt
Sorry, could not understand audio.
PS C:\Users\ABISHAA\Downloads\UID> python vui.py
 Listening... Say: Create file filename.txt
```